

FAGERNES, Norway

WMO: 013670

Lat: 60.9786N

Lon: 9.2239E

Elev: 365

StdP: 97.01

Time Zone: 1.00 (EUC)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-22.8	-20.5	-25.9	0.4	-22.8	-23.4	0.5	-20.4	6.3	-2.1	5.4	-6.8	2.9	320	0.616

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 9.3	(c) 25.4	(d) 14.9	(e) 23.4	(f) 14.1	(g) 21.4	(h) 13.2	(i) 16.6	(j) 22.1	(k) 15.7	(l) 20.6	(m) 14.7	(n) 19.3	(o) 2.3	(p) 130

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
14.7	10.9	18.3	13.7	10.2	17.2	12.7	9.5	16.3	47.8	22.1	45.0	20.5	42.3	19.4	20.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a) 5.3	(b) 4.5	(c) 4.1	(e) -24.8	(f) 27.8	(g) 3.6	(h) 2.0	(i) -27.4	(j) 29.2	(k) -29.5	(l) 30.3	(m) -31.5	(n) 31.5	(o) -34.2	(p) 32.9		
			DB	WB	DB	WB	DB	WB	DB	WB	DB	WB	DB	WB		
			(e) -25.2	(f) 17.8	(g) 3.5	(h) 1.4	(i) -27.8	(j) 18.8	(k) -29.8	(l) 19.6	(m) -31.8	(n) 20.3	(o) -34.4	(p) 21.3		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	3.6	-8.3	-6.7	-2.0	3.4	8.3	12.8	15.7	14.3	9.9	3.8	-1.6	-6.9		
	DBStd	9.43	6.43	6.39	5.16	3.26	3.53	2.88	2.93	2.66	2.96	3.52	4.56	6.61		
	HDD10.0	2796	568	467	373	199	77	7	0	2	36	193	348	525		
	HDD18.3	5389	827	701	632	449	312	167	92	127	252	449	598	783		
	CDD10.0	463	0	0	0	1	23	91	177	135	35	2	0	0		
	CDD18.3	14	0	0	0	0	0	2	11	2	0	0	0	0		
	CDH23.3	173	0	0	0	0	6	32	107	28	0	0	0	0		
	CDH26.7	19	0	0	0	0	0	3	14	2	0	0	0	0		
(14) Wind	WSAvg	1.8	1.6	1.7	1.9	1.8	1.9	1.9	1.8	1.7	1.7	1.7	1.8	2.0		
(15) Precipitation	PrecAvg	608	42	27	26	31	47	74	85	87	53	55	47	34		
	PrecMax	828	85	56	59	76	148	152	185	162	94	143	141	59		
	PrecMin	411	10	5	4	3	0	8	25	12	21	8	11	5		
	PrecStd	89	16	14	14	18	30	33	41	39	18	31	27	15		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	7.0	8.3	11.3	16.6	24.1	26.5	28.4	26.0	21.1	14.9	10.1	7.7		
		MCWB	3.5	3.8	5.9	8.7	13.2	14.4	16.5	14.8	13.2	10.4	6.6	4.2		
	2%	DB	4.2	5.8	8.8	13.7	20.3	23.6	26.2	23.6	18.1	12.0	7.7	5.1		
		MCWB	1.4	2.2	4.1	6.8	10.8	13.3	15.8	14.5	11.7	8.5	5.1	2.8		
	5%	DB	1.9	3.6	6.7	11.4	17.7	21.3	24.2	21.6	16.4	10.4	5.5	3.2		
		MCWB	0.2	0.9	2.5	5.6	9.7	12.0	14.7	14.0	11.2	7.7	3.8	1.4		
	10%	DB	0.4	1.7	4.8	9.2	15.1	19.0	22.1	19.5	14.8	9.1	3.8	1.5		
		MCWB	-0.7	-0.2	1.2	4.3	8.5	11.1	13.9	13.3	10.6	6.7	2.5	0.3		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	3.8	4.1	6.2	9.5	14.2	15.6	18.6	17.4	14.5	11.3	7.5	4.8		
		MCDB	6.7	7.7	10.8	15.0	21.1	21.5	25.0	22.2	18.3	13.9	9.4	7.1		
	2%	WB	1.7	2.5	4.4	7.4	12.3	14.4	17.1	16.2	13.2	9.4	5.3	3.0		
		MCDB	3.7	5.2	8.4	12.3	17.7	20.9	23.1	20.7	16.3	11.1	7.1	4.7		
	5%	WB	0.4	1.2	2.8	6.1	10.8	13.5	16.2	15.3	12.3	8.4	4.0	1.5		
		MCDB	1.6	3.3	6.1	10.1	15.8	19.1	21.8	19.4	15.2	10.0	5.3	2.8		
	10%	WB	-0.7	0.0	1.6	5.0	9.4	12.4	15.2	14.3	11.3	7.0	2.6	0.4		
		MCDB	0.4	1.5	4.2	8.6	14.3	17.2	20.2	17.9	13.9	8.8	3.7	1.4		
(35) Mean Daily Temperature Range	5% DB	MDBR	6.0	7.3	9.2	9.2	9.9	9.5	9.3	8.7	7.8	5.6	4.5	5.4		
		MCDBR	7.8	8.7	10.2	13.2	15.4	14.4	13.2	12.2	10.4	7.4	6.8	7.5		
		MCWBR	6.2	6.2	6.4	7.1	7.1	6.0	5.0	4.7	5.5	4.8	5.3	5.8		
		MCDBR	7.2	8.2	9.8	11.8	12.3	10.9	10.8	9.7	8.5	6.2	6.2	7.0		
	5% WB	MCWBR	5.9	6.1	6.4	6.8	6.3	5.3	4.7	4.8	5.1	4.6	5.1	5.7		
(40) Clear-Sky Solar Irradiance	taub		0.219	0.265	0.300	0.339	0.346	0.342	0.357	0.348	0.314	0.297	0.236	0.187		
	taud		2.137	2.248	2.229	2.241	2.394	2.462	2.434	2.448	2.530	2.466	2.205	2.016		
	Ebn at Noon		545	732	819	852	878	888	862	840	816	699	513	406		
	Edh at Noon		37	64	92	111	104	99	100	91	70	53	34	24		
(44) All-Sky Solar Radiation	RadAvg		0.31	0.98	2.48	4.23	4.91	5.12	4.65	3.64	2.37	1.10	0.38	0.18		
	RadStd		0.06	0.20	0.24	0.51	0.53	0.50	0.56	0.30	0.28	0.18	0.07	0.01		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	+0.72	N/A	N/A	N/A	N/A
Regional (56 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page